Remarks on Bedert's "On unique sums in Abelian groups"

Mark Watson

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This note collects corrections, sharpenings, finite-group results, and data related to Bedert's paper [B]. Parts II and III contain observations obtained in the course of checking the argument. They are offered as supplements to the paper, not as claims about what its author intended to optimize. For a finite subset A of an Abelian group, [B] calls $a_1 + a_2$ a unique sum when its only ordered representations are (a_1, a_2) and (a_2, a_1).

Part I
Corrections and scope

1 A 2-torsion scope issue

At [B, line 291], the proof says that a set with no unique sum is balanced: the sum $a + a$ has another representation, which is taken to have distinct summands. In a group with 2-torsion, however, the other representation may be another diagonal pair. Consider

$$A = \{0, 1, 2, 3, 5, 6, 8\} \subset \mathbb{Z}/12\mathbb{Z}.$$

The numbers of unordered representations of the twelve residues are

$$(2, 2, 3, 2, 3, 2, 3, 2, 3, 2, 2, 2).$$

Thus A has no unique sum under [B, line 62]. It is not balanced: neither 0 nor 6 is the midpoint of two distinct members of A. The proof of the general-group lower-bound theorem invokes "no unique sum implies balanced" at [B, line 458]. The printed proof therefore needs an odd-order clause at that step, or a separate argument for groups with 2-torsion. This does not dispute the theorem itself.

The boundary is sharp among cyclic groups of even order.

Theorem. *For $m \geq 0$, there is a set $A \subset \mathbb{Z}/(2m)\mathbb{Z}$ with no unique sum and some ordered representation count equal to 2 if and only if $m \geq 6$.*

For $m \geq 6$, one may take

$$\{0, \ldots, m\} \setminus \{m - 2\} \ \cup \ \{m + 2\}.$$

Verification: kernel-certified. The family and the exhaustive cases $2m < 12$ are checked in `CycEven.lean`; `verify.py` recomputes the C_{12} table.

2 The Łuczak–Schoen upper bound

Łuczak and Schoen [LS, Theorem 2] use the ordered representation count of [B]. Taking $Q = 2$ gives, for $p \geq 2^{22}$, a set $A \subset \mathbb{Z}/p\mathbb{Z}$ with

$$|A| \leq 4(\log_2 p)^2, \qquad r_A(g) \geq 4 \quad (g \in A + A).$$

Hence $m(p) \leq 4(\log_2 p)^2$. Thus the order of growth in [B, Theorem 5] already follows from [LS]. Bedert's construction improves the constant and is more direct. The author has confirmed this observation and intends to add a note.

3 A redundant factor in Proposition 1

For an indexed family $Z = (z_1, \dots, z_n)$, let $\Sigma(Z)$ be its set of subset sums and let d be the largest size of a dissociated index set. Then

$$|\Sigma(Z)| \le \#\{I \subseteq [n] : I \text{ is dissociated}\} \le \sum_{j=0}^{d} \binom{n}{j} \le \binom{n+d}{d}.$$

Choose the colexicographically least index set representing each subset sum. Every set shattered by this family is dissociated, and Pajor's strengthening of Sauer–Shelah completes the count. This removes the first factor from $\binom{n}{d}\binom{n+d}{d}$ in [B, Proposition 1].

Verification: kernel-certified first two inequalities; finite arithmetic check for the last. `SubsetSums.lean` checks the first two inequalities and `verify.py` checks the Vandermonde step. The author has explained that he chose the simpler sufficient bound because the factor does not affect the asymptotic theorem.

4 The empty set

The definition at [B, line 282] makes the empty set balanced, while Proposition 3 asks for $g \in -B$ for every balanced B. The repair is to require balanced sets to be nonempty in the definition, or in the propositions that use such an element.

Part II
The general-group question

5 The minima three and four

Theorem. *For every finite Abelian group G,*

$$m(G) = 3 \iff 3 \mid |G|.$$

Moreover, if $4 \mid |G|$ and $3 \nmid |G|$, then $m(G) = 4$.

Proof. Translate a three-set to $\{0, x, y\}$. If it has no unique sum, the pair $\{0, x\}$ can collide only with $\{y, y\}$, so $x = 2y$. Symmetrically $y = 2x$, whence $3x = 0$. Conversely, a coset of a subgroup of order three has no unique sum. This proves the first assertion by Cauchy's theorem.

If $4 \mid |G|$, choose a subgroup H of order four. For $a, b \in H$, choose $h \in H \setminus \{0, b - a\}$. Then

$$(a + h) + (b - h) = a + b$$

is neither the original nor the swapped pair. Thus H has no unique sum. The first assertion and the elementary two-point obstruction give the lower bound when $3 \nmid |G|$. ☐

Verification: proved, elementary; algebraic core kernel-certified. `Dcyc.lean` checks the three-set algebra, both order-four groups, and the necessary C_{12} counterexample to the unguarded four-branch.

6 Census-52

Verification: exact computation, DRAT-certified. All 88 Abelian groups of orders 2 through 52, and 26 separately labelled stress groups through order 221, match

$$m(G) = \min\big(\{4 : 4 \mid |G|\} \cup \{m(p) : p \text{ odd prime},\ p \mid |G|\}\big), \tag{2}$$

with C_2 separately having no nonempty such set. This is 114 exact groups, not a theorem for all finite Abelian groups. Each of the 114 lower bounds has a retained DRAT proof and verified checker log; the 113 upper witnesses were recounted directly. The proofs are not distributed with this note. Their CNF and proof SHA-256 hashes, paths, group sizes, and outcomes are recorded in `mg-census-through-52.json`, whose own SHA-256 is

`f8d1c7119e5e98f6d97c7bd4bc76e7eeac1060915876196040f6df74f9a073e9`.

The manifest does not record proof byte counts, so no byte-size claim is made here.

7 A double-cyclic conjecture

Conjecture. *For every odd prime p,*

$$m(C_{2p}) = m(C_p). \tag{3}$$

The following evidence and reduction do not prove (3).

First, a DRAT-certified exact search shows that every minimum set in $C_2 \times C_p$ lies in one parity fibre for $p = 3, 5, 7, 11, 13, 17, 19, 23$. Second, write

$$A_e = \{x : (e, x) \in A\}, \quad B = A_0 \cup A_1, \quad D = A_0 \cap A_1.$$

If A is minimum and $D = \varnothing$, projection is injective and B has no unique sum. Hence $m(C_p) \le |B| = |A|$, while embedding a minimum set of C_p in one fibre gives the reverse inequality.

Verification: proved reduction. Thus empty overlap at the minimum implies (3). Eliminating nonempty residual overlap remains open.

The naive statement that projection preserves the property is false. In $C_2 \times C_{11}$, the full double lift of

$$\{0, 1, 2, 4, 7\} \subset C_{11}$$

has no unique sum, although its projection does. This counterexample is kernel-certified in `Dcyc.lean`. It is not minimum, so it does not refute (3).

Part III
Sharpenings of the prime-cyclic argument

8 A one-log improvement

Put

$$M = \log_2 \log_2 p, \qquad K(A) = \frac{|A|}{\dim A}.$$

Replacing the update $|S'| \le \max(2|S|, |S|^3)$ in [B, lines 479–494] by a two-point update throughout gives the following conclusion under the hypotheses displayed in the formal statement:

$$K(A) > \sqrt{M/404}. \tag{1}$$

This is a genuine asymptotic improvement, replacing the triple logarithm under the square root by a double logarithm. It is numerically vacuous at every computable prime: the free inequality $K(A) \ge 1$ dominates (1) below

$$p > 2^{2^{404}}.$$

Both facts are part of the claim.

There are two formal statements. The statement in `ObjectLayer/O7Headline.lean` is

Verification: conditional, kernel-certified; frontier hypothesis. conditional on the stated Lev rectification interface `hBLR`. In that formalization we do not establish this hypothesis, and nothing in the development supplies an instance of it. The extracted theorem `sqrt_improvement_from_isUSF_internal` has the identical remaining hypotheses and conclusion but no `hBLR` binder.

Verification: kernel-certified, unconditional. Its 19-module import closure is included with this note and prints exactly `[propext, Classical.choice, Quot.sound]`. The associated satisfiability check supplies a concrete model of the arithmetic hypotheses, so the unconditional statement is not vacuous merely by inconsistent numeric assumptions. The paper itself says that the simplest sufficient form of this slowly growing factor was chosen and that its exact shape was not optimized [B, after Theorem 3].

9 Sharpness of the simultaneous-selector threshold

Let $f(p)$ be the largest cardinality for which the selector conclusion used in [B, Lemma 5.1] holds for every $S \subset \mathbb{Z}/p\mathbb{Z}$. Then

$$f(p) = \lceil \log_2 p \rceil.$$

Equivalently, a selector exists for every k-set precisely while $2^{k-1} < p$. The general upper obstruction is

$$\{0\} \cup \{\lceil p/2^i \rceil : 1 \leq i \leq \lceil \log_2 p \rceil\}.$$

Verification: exhaustive computation at 29 primes across five complete bands, plus a hand proof of the general upper bound with machine-checked algebraic identities. The exact computations cover all primes in the five bands from $(4, 8)$ through $(64, 128)$. In particular, all 234,531,275 seven-subsets pass at $p = 127$, while an eight-subset fails. The rectification threshold used here originates in Lev's Theorem 1 [L]. The gain over the displayed $2^{|S|} \leq p$ hypothesis is exactly one element, not a change of scale.

Part IV

Data

10 Exact values of $m(p)$

p	3	5	7	11	13	17	19	23	29	31	37	41	43	47	53	59
$m(p)$	3	4	5	7	7	8	9	10	11	11	12	13	13	13	14	15

p	verification tier
$3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37$	exact solver output, independently corroborated; not certified here
$41, 43, 47$	DRAT-certified optimality and verified witness
$53, 59$	exact by exhaustion and verified witness; not DRAT-certified

The requested $p = 53$ cloud metadata object was not present when this version was staged, so its tier has not been upgraded. The checking script redoes the small cases feasible in pure Python and verifies the displayed witnesses at $41, 43, 47, 53, 59$.

Checking

Run

`lake exe cache get && lake build` and `python3 verify.py`.

The principal Lean axiom prints should contain exactly `propext`, `Classical.choice`, and `Quot.sound`.

References

[B] B. Bedert, *On unique sums in Abelian groups*, Combinatorica **44**(2) (2024), 269–298. DOI: 10.1007/s00493-023-00069-w; arXiv: 2303.15134.

[L] V. F. Lev, *The rectifiability threshold in Abelian groups*, Combinatorica **28** (2008), 491–497.

[LS] T. Łuczak and T. Schoen, *On a problem of Konyagin*, Acta Arith. **134**(2) (2008), 101–109.

[P] A. Pajor, *Sous-espaces l_1^n des espaces de Banach*, Travaux en Cours 16, Hermann, Paris, 1985.